1. Highest Relevance: Competency #12 — Use a multi-step procedure leading to a mathematical solution

- Match: Solving requires a sequence of logical steps:
  (1) Recognize the geometric model (right triangle);
  (2) Identify known values (opposite side = 150 m, angle = 45°) and unknown (hypotenuse);
  (3) Select the appropriate trigonometric ratio (sine);
  (4) Perform the calculation;
  (5) Approximate and match to the closest option.
- Analysis: This is not a single-step task but demands coherent procedural reasoning.
- ✅ Relevance: Very High

2. High Relevance: Competency #6 — Develop mathematical diagrams and extract mathematical information from them

- Match: The problem includes a schematic diagram (labeled "not to scale"), from which students must infer key geometric relationships: vertical leg, 45° angle, and rope as hypotenuse.
- Analysis: Even though the diagram isn't precise, it conveys essential structural information necessary for modeling.
- ✅ Relevance: High

3. High Relevance: Competency #13 — Use an understanding of the context to guide or expedite the mathematical solving process

- Match: Students must interpret real-world phrases like "vertical height of 150 m" and "pulling at a 45° angle" as geometric elements in space.
- Analysis: Without contextual understanding, constructing the correct triangle model would be impossible.
- ✅ Relevance: High

4. Moderate Relevance: Competency #7 — Manipulate numbers and geometric representations

- Match: Involves numerical computation ($150 \div \sin 45^\circ$) and handling geometric representation (right triangle).
- Analysis: The manipulation is straightforward but essential to the solution process.
- ✅ Relevance: Moderate

5. Moderate Relevance: Competency #5 — Understand and utilize constructs based on definitions, rules, and formal systems

- Match: Requires correct application of the definition of the sine function ($\sin\theta = \text{opposite}/\text{hypotenuse}$).
- Analysis: Relies on conceptual understanding of trigonometric ratios, not rote memorization.
- ✅ Relevance: Moderate

1. 最高度相关：第12项 — Use a multi-step procedure leading to a mathematical solution

- 匹配点：解题需多个逻辑步骤：
  (1) 识别几何模型（直角三角形）；
  (2) 确定已知量（对边、角度）和未知量（斜边）；
  (3) 选择合适的三角函数（sin）；
  (4) 代入计算；
  (5) 近似取值并匹配选项。
- 分析：这不是单步计算，而是需要连贯的推理链条。
- ✅ 相关性：极高

2. 高度相关：第6项 — Develop mathematical diagrams... and extracting mathematical information from them

- 匹配点：题目提供了一个简图（尽管“not to scale”），学生需从中提取关键几何关系：垂直高度、角度、绳索作为斜边。
- 分析：即使图不精确，它仍传达了结构信息（直角、45°角、150m垂直边），这是建模的基础。
- ✅ 相关性：高

3. 高度相关：第13项 — Use an understanding of the context to guide or expedite the mathematical solving process

- 匹配点：学生需理解“vertical height of 150 m”和“pull at an angle of 45°”在物理空间中的含义，并将其转化为几何元素。
- 分析：脱离情境就无法建立正确的三角形模型。
- ✅ 相关性：高

4. 中等相关：第7项 — Manipulate numbers... and geometric representations

- 匹配点：涉及数值计算（150 ÷ sin45°）和几何表示（直角三角形）。
- 分析：操作本身不复杂，但属于解题必要环节。
- ✅ 相关性：中等

5. 中等相关：第5项 — Understand and utilise constructs based on definitions, rules and formal systems...

- 匹配点：必须正确应用三角函数的定义（sin = 对边/斜边）。
- 分析：依赖对三角学基本规则的理解，而非死记硬背。
- ✅ 相关性：中等